RESUME

Ignacio Cutignola

Buenos Aires, Argentina

Firmware & Embedded Systems Developer · Electronics Engineering Student (ITBA)

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PROFILE

Electronics Engineering student at ITBA (85% of the degree completed) and Electronics Technician. I currently develop firmware and hardware for embedded systems at Prosegur, with previous experience in cybersecurity (Accenture) and full-stack development (Balanz). As Director and Tutor at the IEEE ITBA Student Branch, I designed and taught programming courses that reached 430 students. I am looking to keep growing in firmware, embedded systems, and software development roles.

WORK EXPERIENCE

Semi-Senior Project Analyst | Prosegur

02/2026 - Present

- Firmware development for alarm panels, sensors, access control, and IoT devices on AVR (ATmega) and ESP32/ESP8266 microcontrollers.
- PCB design in Altium Designer and integration of UART, SPI, I2C, RS-485/Modbus, Wi-Fi, and Ethernet communications.
- Testing, validation, and commissioning of devices, providing technical support during installation.
- Supervision of a technical team of up to 3 people and management of development projects.

Tools: Altium Designer, Atmel Studio, Arduino, Python, BASCOM, C, C++, Fusion 360

Junior Cybersecurity Analyst | Accenture

07/2025 - 01/2026

- Cybersecurity incident response for telecommunications networks.
- Detection and analysis of alerts and events to determine their scope and impact.
- Incident documentation and preparation of technical reports.
- Work in Linux environments (Kali Linux and Ubuntu).

Tools: Kali Linux, Ubuntu

Junior Full-Stack Developer | Balanz

01/2022 - 07/2023

- Full-stack development of the investment platform, internal back-office systems, and API integrations using Node.js, Python 3, Angular, and TypeScript.
- Management of SQL, MongoDB, and Redis databases, improving query and service performance.
- Automation of internal processes and reports, and development of new features alongside business teams.
- Production support: diagnosing and fixing recurring incidents.
- Agile teamwork (Scrum) using Git and API testing with Postman.

Tools: Node.js, Python, Angular, TypeScript, SQL (PostgreSQL), MongoDB, Redis, Postman

Director and Tutor | IEEE ITBA Student Branch

03/2020 - 07/2023

- Designed and taught introductory and advanced courses on Python 3, TypeScript, and IoT, reaching 430 students during the pandemic.
- Created the syllabus, study materials, and exercises for each course.
- Coordinated a team of up to 10 volunteer tutors.
- Managed branch operations: enrollment, outreach, and events.

Recorded class: youtube.com/watch?v=AAAT5VprKgg

Tools: TypeScript, Python

Intern | Baxo S.A.

12/2019 - 12/2020

- Configuration of routers, switches, and wireless networks (Wi-Fi and point-to-point links).
- Work with fiber optics, structured cabling, and IP telephony (VoIP).
- Technical support for clients: diagnosing and resolving network failures.

EDUCATION

Electronics Engineering | Instituto Tecnológico de Buenos Aires

2017 - Present

85% of the degree completed

Electronics Technician | Escuela Cristiana Evangélica Argentina

2011 - 2016

TECHNICAL SKILLS

Embedded & Hardware: C, C++, Assembly, Atmel Studio, BASCOM, Arduino, Raspberry Pi, Altium Designer, LTspice, EPLAN

Software & Data: Python, TypeScript, Node.js, Angular, SQL (PostgreSQL), Redis, MongoDB, Postman

Systems & Security: Linux, Ubuntu, Kali Linux

Design, Computing & Documentation: Fusion 360, Blender, MATLAB, LaTeX, Photoshop, Premiere Pro, Excel

LANGUAGES

Spanish: Native **English:** Upper-Intermediate (B2) **Portuguese:** Intermediate

HONORS & AWARDS

ITBA Honors Scholarship

PROJECTS

Embedded MP3 player

Academic project

C firmware for an MP3 player on the NXP FRDM-K64F board (ARM Cortex-M4F at 120 MHz) running the µC/OS-III RTOS. It reads files from a micro SD card with FatFs, decodes them with the Helix library, and plays the audio through the 12-bit DAC using DMA with a ping-pong buffer, timed by the PIT at the sample rate. It includes a 4-band equalizer built with CMSIS-DSP biquads, ID3v2 metadata reading, volume control, a user interface with a 16x2 LCD, rotary encoder, and push buttons, a VU meter on a WS2812 LED matrix, and a custom PCB designed in Altium Designer.

GitHub repository: github.com/cutignolai/MP3-Player

Tools: C, Altium Designer

RTOS-based access control

Academic project

C firmware for an access control system on the NXP FRDM-K64F board (ARM Cortex-M4) running the µC/OS-III RTOS, migrated from a bare-metal polling version to an architecture of tasks, semaphores, a message queue, and events handled with OSPendMulti. Users identify themselves with an 8-digit ID, entered with a rotary encoder or read from a magnetic stripe card (track 2, ISO/IEC 7811, with parity and LRC validation), and then enter a PIN. It includes custom drivers for interrupt-driven GPIO, software timers, a quadrature-decoding encoder, a push button with multi-click and long-press detection, a multiplexed 7-segment display with 3 brightness levels, and status LEDs, plus a state machine with time penalties and lockout on the third failed attempt.

GitHub repository: github.com/cutignolai/Control-de-acceso-TP4

Tools: C

Color classifier

Academic project

Team-built machine that sorts products by color onto a rotating disc, with automatic and manual modes. It integrates a DC-motor conveyor belt, a color sensor, an inductive sensor for homing, infrared sensors for detection and arrival, a stepper motor, Arduino Nano control, an operator safety panel, electrical panel drawings designed in EPLAN, and 3D-designed and modeled parts.

LinkedIn post: lnkd.in/p/dekPENSN

Tools: Arduino, C, Altium Designer, EPLAN, Fusion 360

Signal sampling and recovery simulator

Academic project

Python desktop application (PyQt5, NumPy, SciPy, and Matplotlib) that simulates the full sampling and reconstruction chain of an analog signal. It models an 8th-order anti-aliasing filter (Legendre approximation) built from 4 Sallen-Key cells defined by real components, a sample & hold, an analog switch with configurable duty cycle, and a recovery filter. Each stage can be enabled or bypassed and is displayed in time and frequency (FFT), on an extensible architecture with a common base class.

GitHub repository: github.com/cutignolai/ASSD-TP1

Tools: Python

Robotic arm that draws from images

Academic project

Together with my university team, we built a robotic arm that receives an image of a drawing, interprets it, and replicates it.

Tools: MATLAB, Computer Vision

Pulp Vision 3D

Personal venture

3D printing venture: design and manufacturing of mechanical parts and custom spare parts on an Ender 3 SE, in PLA and PETG, from modeling to sale.

Tools: Fusion 360, Blender, Photoshop, Premiere Pro, Excel

Home stock manager

Personal project

Home system that registers products with a barcode reader and uses a Raspberry Pi as its central brain: it tracks stock, sends email alerts when something runs out, and builds the shopping list.

Tools: Arduino, Raspberry Pi, MongoDB, SQL (PostgreSQL), Linux

Private lessons

Private tutoring

Lessons in Physics, Mathematics, C/C++, and Python 3 for a group of 5 to 10 students, with a focus on conceptual clarity and personalized support.

Tools: Python, C++